

Does the Handover Margin Earn Its Place?

A Factorial Comparison of Temporal Decision Policies for Context-Based Continuous Smartphone Authentication

Finding: the margin did not earn its place - a short dwell captured most of the measurable stabilisation, at lower cost.

Postgraduate Final Viva Presentation
Computer Science / Data Science Project Work

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Education

Presentation Outline

Background: why a score stream is not an authentication state

- Aim, objectives and the research question
- Related work and the identified gap
- Methodology: frozen score stream, calibration-only tuning, preregistration
- Design of the decision policies, and what was implemented
- Results: the preregistered test, and evaluation across nine mechanisms
- Contribution, limitations, conclusion and future work

Background and Problem Context

- Continuous authentication keeps confidence in the current phone user after login, using passive signals.
- A classifier emits a score every frame; a device must hold a persistent state - authenticated, or locked.
- Near the threshold that state flickers even when the user has not changed. AUC cannot detect it: it ranks frames and ignores their order.



Instantaneous threshold

8 state changes



With a persistence requirement

1 state change

Ribbons are an illustrative schematic, not measured data. Filled = authenticated. Real traces appear on slide 7.

Problem statement: converting a noisy score stream into a stable authentication state is a design layer in its own right, and frame-level accuracy cannot measure it.



Aim, Objectives & Questions

- **Aim: to test whether a handover-inspired margin adds useful stabilisation beyond temporal dwell, at a common calibration-defined operating point.**
- **Objectives**
 - Decompose margin and dwell on one frozen score stream under equal tuning.
 - Select every operating point from calibration data only, at a decision-layer FAR target of 0.05.
 - Evaluate at participant level on stability, FAR, FRR, detection and recovery - not frame accuracy alone.
 - Test the construct validity of the score itself.

Research question

At a calibration-defined decision-layer FAR target, does adding a handover-inspired margin to temporal dwell reduce excess authentication-state transitions without unacceptable false rejection or detection failure?

Scope: one public dataset, one score generator, offline replay. Excluded - classifier design, live deployment, and any claim about biometric identity.



Related Work and Identified Gap

- Zeeshan et al. (2025) use EWMA "to stabilize decisions and avoid flickering" - but never isolate its effect, report the coefficient, or test a margin.
- Mondal & Bours (2015) do compare decision layers, but sweep fusion, thresholds, boosting and trust models together, reporting action-domain measures.
- Sugrim et al. (2019) show equal-error-rate comparison is uninformative when a system has a fixed false-accept requirement; compare at a fixed operating point.
- Stragapede et al. (2023) show mobile behavioural models can identify the device rather than the user.

Identified gap

Prior work compares temporal-policy combinations. Evidence is limited on the incremental effect of a dual-threshold margin when margin and dwell are decomposed under equal tuning, on a shared score stream, at a calibration-defined FAR target.

The novelty claimed is the decomposition - not the classifier, the dataset, or temporal smoothing itself.



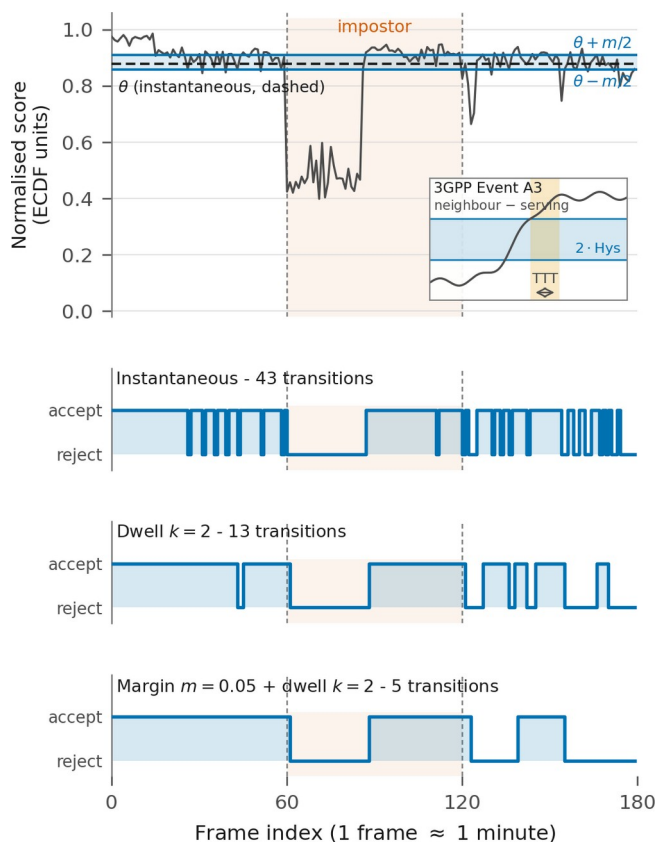
Methodology / Project Approach

- Experimental reanalysis of the public ExtraSensory dataset (Vaizman et al., 2017) - collected for context recognition, not authentication.
- Cohort: 31 participants on a regular ~1-minute cadence, defined before any confirmatory score was generated. Each is one enrolled user, one-versus-rest.
- Chronological enrolment / calibration / test partitions. Preprocessing fitted on enrolment only. Final-test impostors unseen during fitting and calibration.
- Per-user gradient-boosting score generator, frozen before any decision policy is applied. Every policy replays the same stored scores.
- Parameters chosen on calibration data only, at a decision-layer FAR target of 0.05 (preregistered feasible band 0.04-0.05).
- Participant-level paired inference: Wilcoxon signed-rank, Holm-corrected, with bootstrap confidence intervals.

Reproducibility: rules, grids, outcomes and the verdict wording were frozen before the score dumps existed. Executed code is byte-identical to the frozen code. Public dataset, so no further ethical approval was required.



System / Model Design



Decision rule

Accept if score $\geq \theta + m/2$; reject if score $< \theta - m/2$. k consecutive qualifying frames flip the state; anything else resets the counter.

Factorial: $m \in \{0, 0.05, 0.10, 0.20\} \times k \in \{1, 2, 3, 5, 10\}$

Instantaneous

$m = 0, k = 1$

Dwell only

$m = 0, k > 1$

Margin only

$m > 0, k = 1$

Margin + dwell

$m > 0, k > 1$

Evaluation contract

Each test sequence is 60 genuine, 60 unseen-impostor and 60 recovery frames. A correct trace needs exactly two state changes; anything beyond that is an excess transition.

Figure 5. One real test sequence under three policies, on an identical score stream.

Implementation Work Done

Built and executed

- Python, scikit-learn 1.6.1, NumPy, pandas. Score dumps generated from commit f42f55c with seed 20260918.
- Per-frame scores dumped once to Parquet, so every rule change is replayed offline - model fitting was 92% of runtime.
- All 20 factorial cells run per participant (3 infeasible at the target), plus nine further mechanisms for the secondary comparison.

Challenge resolved

A windowed-mean defect made one secondary mechanism disagree with its reference at exact ties. Fixed, regression-tested, and the offline analysis rerun; the preregistered primary test was unaffected.

Preregistration, frozen analysis, figure and table scripts, checks and rendered outputs are all in the project repository.

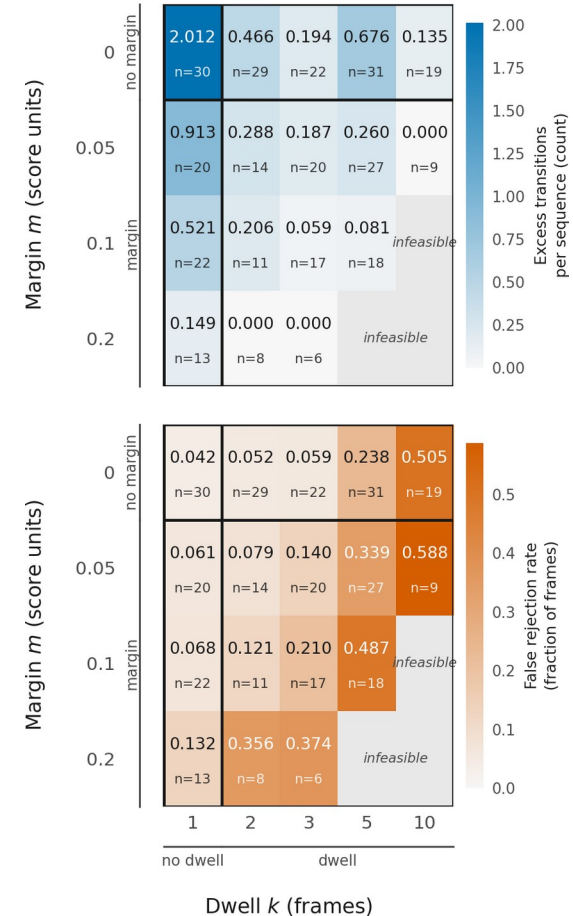


Figure 1. The executed factorial at FAR 0.05: excess transitions (top) and FRR (bottom).



Results and Evaluation

	vs dwell only	vs margin only
Participants paired	30	24
Excess transitions, mean paired difference	-0.13	-0.20
Holm-adjusted p	0.69	0.69
FRR difference	+0.074	+0.030
FRR 95% CI	[0.032, 0.119]	[-0.008, 0.069]
Detection-failure difference	0.000	-0.007

Margin plus dwell offers no measurable advantage over its components, and is FRR-inferior to dwell alone.

- The +0.074 FRR penalty breaches the preregistered +0.02 non-inferiority bound. 20 of 30 participants tied exactly on stability.
- Descriptively, a dwell of 2 frames cuts excess transitions from 2.012 to 0.466 per sequence for about +0.01 FRR (unpaired cells).
- Across nine mechanisms none dominates: the most stable are the most locked out. Ordering holds at FAR 0.03 and 0.07.

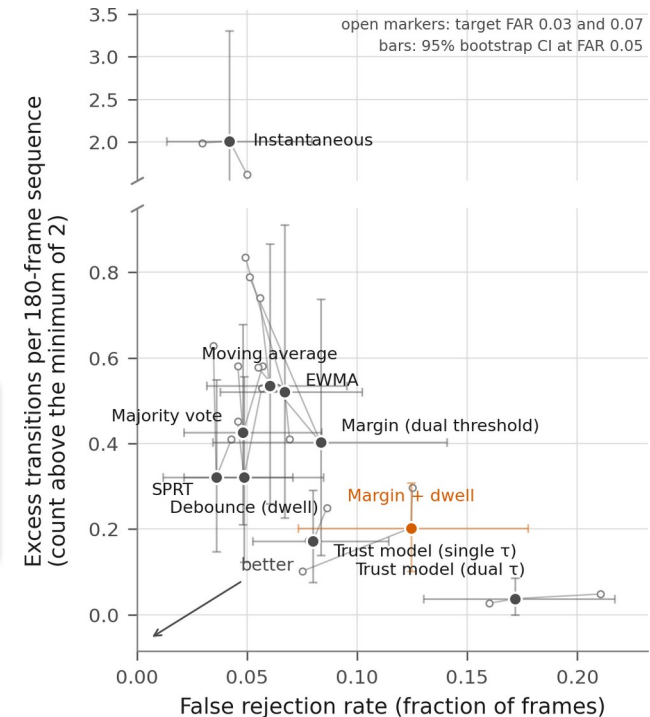


Figure 2. Stability-lockout plane. Lower left is better; no mechanism reaches it.



Contribution and Significance

Evidence

A preregistered null plus a directional failure: no stability gain from the margin ($p = 0.69$ against both components), and an FRR penalty of $+0.074$ against dwell that breaches the $+0.02$ bound.

Interpretation

Persistence is what transfers from the handover analogy. The margin does not add measurable value here - it converts instability into lockout.

Significance

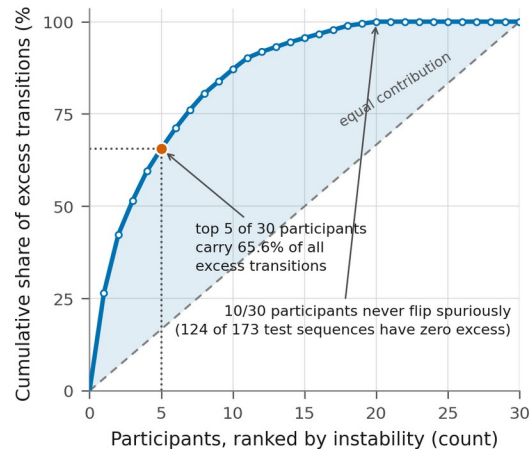
Methodological: a leakage-aware, calibration-frozen way to test whether a decision rule improves a system, rather than merely making its state harder to change.

Practical

For a designer with a comparable score stream, a short dwell is the simplest defensible first intervention. Always report excess transitions together with FAR, FRR and recovery - a stability number alone rewards a policy that has stopped responding.

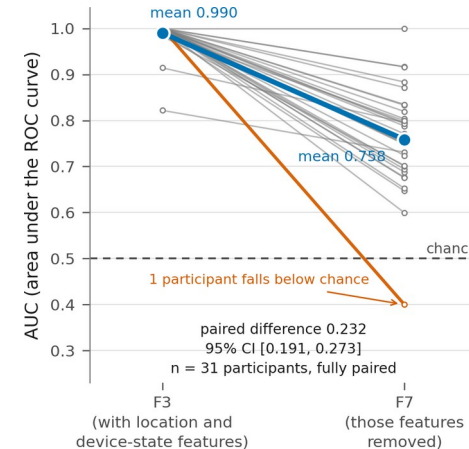


Conclusion and Future Work



Instability is rare and concentrated

71.7% of instantaneous test sequences never flip; 5 of 30 participants carry 65.6% of it. Little for a margin to improve - this limits power.



The score is not clean identity evidence

Removing location and device-state features drops mean AUC from 0.990 to 0.758 (paired difference 0.232 [0.191, 0.273]).

Conclusion: under a preregistered, calibration-frozen comparison, the margin did not earn its place on this benchmark.

Future work: a cohort selected for instability; a dataset that separates user from device; equalised tuning grids so the SPRT result can be confirmed.



References

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APA 7th. Full reference list and the citation verification record are in the dissertation and the project repository.

